Virtual University of Pakistan

ZOO-510

Economic Zoology

| Final term past papers |2019

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ZOO-510

Economic Zoology

M.C.Q 1st Quiz

1. Male Lac-insect; the body is typically divided into head, **thorax** and abdomen.
2. Feeds should be stored in a **dry, cool** and well-ventilated area.
3. **Hand feeding** is a common method of feeding fish.
4. ***Bombyx mori*** Produces over **90%** of the world's raw silk.
5. Blood poisoning due to bacteria is called **Septicemia.**
6. ***Aeromonas hydrophila*** Causes disease in fish.
7. **Pebrine** Causes disease in silkworm.
8. To produce **1 kg** of lac resin, around **300,000** insects lose their life
9. Spring Viremia of Carp is a **viral** infection.
10. Bombyx mori is an economically important insects being a primary producer of **silk**.
11. Whirling disease by **Myxobolus cerebralis**
12. Aghani and baisakhi strain are the main corps contributing about **90%** of lac production.
13. **James I** introduced sericulture to **England** and **America**

M.C.Q 2nd Quiz

1. **Enlargement of hock joints Cause:** def. Of niacin, biotin, vit.E & zinc.
2. **Maganese** is needed of egg production & hatchability.
3. **Methionine** is first limiting AAs for egg production.
4. When **miks** is found to be not up to standard, it may be rejected by the creamery.
5. Mesogenic strain (R2B) for the treatment of ranikhet disease useful at the age of **6-8 weeks of age.**
6. Which one is the Lab Animals – **Mice**.
7. Good Biosecurity Practices should reduce exposure form. **All the above.**
8. Staphylococcus bacteria causes. **BUMBLE FOOT**
9. The milk of the **Jersey cow** has the highest butterfat and SNF content of any cow
10. Fat Soluble Vitamin **A,**

M.C.Q mid term

1. **Aquaculture** is farming of aquatic organisms in natural or controlled marine or fresh water environment.
2. RAS stands for ___ **Recirculating aquaculture system**
3. Induction air into water is called ___ **Aeration**
4. Transformation from soft, dusty mixture to compact form is ___ **Pelleting**
5. fungi are ___ **multicellular**
6. Most common feeding method for fishes is ____ **Hand feeding**
7. Bombyx mori produces ____ **Silk**
8. bombyx mori produces ___**90%** percent of the worlds raw silk
9. In **india** Lac insect is known to have two distinct strains: kusumi and rangeeni.
10. The life cycle of lac insect takes about**....6.....** Months
11. Mature male chicken is called. **Roaster**
12. The lac which grows on non-kusum plants is called as ____ **Ranjeem lac**

13. India alone accounts for about ...**70%**.....percent of global lac production.
14. QMP stands for ____ **Queen mandibular pheromone.**
15. **Poultry**____ is a farming of chicken, turkey, duck and goose.
16. Raceways are considered ___ **Flow through system**
17. ____is a silkworm disease **Pebrine**
18. Aghani and baisakhi strain are the main crops contributing about.....**90%.....** Percent of lac production.
19. Lac cultivation is introduced into.... **Thailand**..... From India.
20. A group of chicken is called ……..**flock**
21. Trained staff called 'catchers' catch the birds and put them in special containers called **modules.**
22. Life of worker bee is **28 to 35 days**.
23. Carnivorous fish consume foods with **~50% of protein**.
24. The place where a beekeeper keeps his bees is called an **apiary or a bee yard**
25. Typically **6 inch** pipe is sufficient for ponds less than 10 acres .
26. Whirling disease Caused by protozoan . ***Myxobolus cerebralis***
27. Mulberry culture spread in India by 140BC from **China**
28. The common parasites of lac insect are known as **"Chalcid"**
29. A virus is a **microscopic organism**
30. American Cancer Association Regular fish consumption decreases chances of **colon cancer 50%**
31. Unsaturated FA = **double bonds**
32. James I introduced sericulture to **England** He also took sericulture to **America**.
33. Female lac insect is **larger** than male.
34. **Septicemia** is the blood poisoning due to bacteria.
35. A parasite which lives inside the host is an **endo parasite**.
36. Spring Viremia of Carp , is a viral infection caused by ***Rhabdovirus carpio***.
37. To produce 1 kg of lac resin, around **300,000 insects** lose their life.
38. ***Aeromonas hydrophila*** caused by fish disease.
39. Feeds should be stored in a **dry, cool and well-ventilated area**.
40. Male Lac-insect: The body is typically divided into head, **thorax** and abdomen.
41. **Ponds** were used as one of the first forms of aquaculture.

M.C.Q final term

1. The hen matures after____**42week**
2. Family of mosquito____ **Mosquitoes** are members of a **family** of nematocerid flies: the **Culicidae** (from the Latin culex, genitive culicis, meaning "midge" or "gnat").
3. The color of York depends upon___ **Caratenoid Pigment**
4. Vitamin D require for_____of rickets

   Bone formation | Egg production | Reproduction and prevention | **All**
5. Vitamin E essential to prevent

   Encyphalomalacia | Crazy chick disease | **Both** | None
6. Riboflavin are also called

   Vit.B2 | Vit.B12 | Vit.B3 | Vit.B
7. Animal protein is good sources of

   Vit.B2 | **Vit.B12** | Vit.B3 | Vit.B
8. Bird drink about _____ as much water by weight of feed consumed

   Twice | Half | Both | None
9. Water consumption ____ according to the environmental temperature

   Low | Increase | Decrease | **Both b and c**
10. Grower ratio is about

    0-3 week | 5-7 week | 1-3 week | **3-5 week**

11. Feed conservation of broilers are____

2.1 **2.2** 2.3 2.4

12. Factor affecting feed conservation is____

Type of feed fed Strain of the bird Feed wastage **All**

13. Feeding a diet containing

10% protein 10 % carbohydrates 10 % lipids All

14. ____is the most common grain used in formulating poultry diet

Breed **corn** Rice meat

15. Animal and vegetables fats added in limited amount

5-10% of diet 5-7 % of diet 1-3 % of diet 3-5 % of diet

16. Animal protein sources are _____ in their amino acid than plants.

less variable **More variable** Both None

17. _____Meal is most commonly plant protein source

Soybean Cotton seed Fish None

18. Fish meal has a good balance of

AAl AAB **AAs** None

19. Na & Cl adding as common salt

0.5-1% of diet 1.5-1% of diet 0.5-2% of diet D-0.5-3% of diet

20. Commercial feeds for poultry as

Mash pellets Crumbles **All**

21. A laying ratio should contain about_____ protein based on 29001 kcal ME/KG of diet

10% 12% 13% **15%**

22. _____ is the first limiting AAs for egg production

Leucine Lysine Arginine **Methionine**

23. Mesh for laying hens should contain not less than ---------- animal protein supplement

3-4% 3-5% 3-0% 2-5%

24. Mineral requirements are

Calcium Salt Phosphorus **All**

25. Bird stored Ca for about ---------- before the first egg was laid in the marrow of long bone

10-14 days 12-14 days 13-14 days 15-16 days

26. Plant protein supplement should supplement with

Na P Ca **Both B & C**

27. Inorganic **P** is ___ than phytate **P**

More available Less available Both None

28. Egg production hen usually covers a period of ____

10 month 12 month 13 month **15 month**

29. The quality of egg shells depend on the presence of adequate levels of vit.D3 & certain mineral including____

Ca P Zn **ALL**

30. The blood carbonate is the source of ____ in the shell formation

Amino acid Carbohydrates Lipids **Carbonate**

31. color of egg yolk depend upon the presence of _____ in the ratio

Carotenoid pigment Xanthophyll's **Both** None

32. Nutritional disorder of turkey is

10 9 8 **7**

33. Causes of leg weakness disorder are

def. Of **Ca** def. Of **P** def.of choline **All**

34. Geese will start to eat pasture when they are only

Few days ago One days ago Two days ago Three days ago

35. There are basically _____ types of disease that must be considered in poultry production.

One or two | One | **Two** | Three

36. Causes of health issues are____

Five | One | Two | Three

37. _____is the cheapest, most effective means of disease control available

Vector | Disease | **Biosecurity** | None

38. Major components of biosecurity are

Flock management | Isolation and traffic control | Cleaning/sanitation | **All**

39. _____ are very effective at killing many disease causing organism

Drying | Sunlight | **Both A and B** | None

40. _____steps to cleaning disinfection (C&D)

Two | Three | Four | **Five**

41. What is disease_____?

Any condition that results in deviation from normal function

42. Infectious agents are____

Bacteria | Viruses | Parasites | Fungi | **All**

43. Non -Infectious agents are_____

Chemical | Physical | Lack or excess of certain vitamin and minerals | Toxins | **All**

44. . Fowl pox vaccine at _____ weeks of age

4-6 | 12-14 | 4-8 | 12-16

45. Riboflavin @ 3.6 of feed in_____

layers | Growers | **Chicks** | None

46. A female calve (destined for dairying) weigh approx_____ at birth

10kg | 20kg | 30kg | **40kg**

47. The length of oestrous cycle of cow is _____days and it latest an average of 18 hours

21 days | 22 days | 23 days | 24 days

48. The gestation period of the dairy cow is _____ (Approx. 9.5 month)

282 days | 281 days | 282 days | **283 days**

49. The yield of average dairy cow decreases after the _____ lactation

4th | **5th** | 6th | 7th

50. The most intervals are _____ hour intervals

14/10 | 13/11 | **Both A and B** | None

51. The milk at the start may only contain ____ fat while the milk at the end (stripping). May contain ___ fat

1% / 10% | 2% / 10% | 3% / 10% | All

52. Dirty milk contain_____

Dirt | Bacteria | Antibiotics | **All**

53. Dirty milk is caused by_____

Dirty milk is caused by | Poor milk filters | **Both A and B** | None of these

54. _____ will also be affected by soil type

Vitamins | Amino acid content | Minerals | **All**

55. Examples of monogastric (Simple stomach) are____

Pig | Dog | Human | **All**

56. Essential nutrients are____

Water | Carbohydrates | Fat protein | Minerals | Vitamins | **All**

57. Ruminants have _____ compartment or stomach

2 | 3 | **4** | 5

58. The total amount of feed an animal has in a _____ period

21hr | 22hr | 23hr | **24hr**

59. Water makes up _____ of the weight of an animal body

75% | 78% | 79% | 9%

60. Types of carbohydrates are

Sugars	Starches	Fiber	**All**

61. Fats have _____ time more energy than carbohydrates

2.25	2.26	2.27	2.28

62. In 23 amino acids ____ are essentials

9	**10**	11	12

63. ____ is the most common deficiency

Carbohydrates	**Protein**	Lipids	Nucleic acid

64. Macro minerals are_____

Calcium	Phosphorous	**Both**	None

65. Types of vitamins are_____

1	**2**	3	4

66. Fat soluble vitamins are_____

A, D, E, K	C&B(*Water soluble vitamins*)	Both	None

67. Classes of feed are_____

6	7	**8**	9

68. Carbonaceous silages are_____

Corn	Grass	**Both**	None

69. Characteristics of roughage feed stuff are _____

Generally low in energy	Higher in fiber	Higher mineral content	**All**

70. Small grains are_____

wheat	grains	**Both**	None

71. Corn stover are called **stalklage**

72. Leaves have _____ protein than stem **2-3x**

73. _____ moisture may lead to mold and excess heating **20-30%**

74. Acitic acid first acid reduces ph from _____ in the first 4 days **6 -- 4.2**

75. Lowered ph prevents bacterial growth and further fermentation as long as it is without **oxygen**

76. _____ is most common very difficult to make well **Haylage**

77. Trace minerals are **Copper sulphate**

78. ____is the most supplemented and has the most health implications attached to its (especially in cattle) **E**

79. Water soluble vitamins are **Riboflavin/Niacin/Choline**

80. Types of mixers are **2**

81. What is forage **Vegetables matter in a fresh dried or ensiled state**

82. _____allow you to raise an agricultural crop on land where other crops cannot be produced **Forage**

83. Healthy plants will have a higher **nutritional value**

84. All root growth stops for 12 days with 80% removal and 18 days with ____removal **90%**

85. In the long run the animals will have more ____ to graze **QForage**

86. Plants are often _____ in the system **overgrazed and under grazed**

87. What is pest?

Technically an organism (bacteria, fungi plants and animals) that has a negative effect on human health orecnomics(food)

88. _____ include insect, mites, plant pathogens, weeds, mollusks, fish, birds and mammals **pests**

89. Thousand of years ago, chemicals that affected pest were discovered through **Trail and error**

90. When sulpher was burned it was found to affect **Insect and mites**

91. Modern chemistry was used to develop pesticides during **World war II**

92. However during the ____ it became obvious that relaying completely on pesticides for pests management could have some negative impact **1970'S**

93. A prime example is the _____ which has develop resistance to every major group of insecticides **Colorado potato bettle**

94. Invertebrates include the wide variety of insect like **Aphids, caterpillars, grubs and wasps**
95. Vertebrates pests include all of those animals with a _____ **Backbone**
96. Cold ,heat ,pollutants and dog urine are _____ **Non living agents**
97. **Natural controls** include wind ,temperature extremes, humidity levels and rainfall
98. Naturally occurring predators and pathogens hold down _____ **pest population**
99. _____are bred to be resistance to the attack of pests **Animals or plants**
100. Some time ____ control do not provide sufficient suppression of pest number **Natural**
101. The cane toad was introduced in Australia in ___to control two pests of sugar cane , butlater emerged as invasive species itself **1935**
102. **Mechanical** control involve using devices, machines, or other physical method to reduce pest population
103. ____is one of the most common forms of mechanical control **Cultivation**
104. **Physical control** is the altering of physical environment – changing humidity, temperature, air flow, water and light
105. Each gene may have several variation called **Alleles**
106. Any material that is applied to kill, attract, repel or regulate pest called _____ **Pesticides**
107. **Contact** pesticides must come in direct contact with the target pest
108. **Glyphosate** is non selective when applied to green plants
109. **Persistence** is the term used for how long a pesticides remain present in the environment
110. **DDT** which is no longer used was very presistant insticides that still found in the environment today
111. **Quarantines** are used at airports and ocean ports to prevent entry of foreign pest
112. **Occasional pest** may become troublesome from time to time
113. **Secondary pest** become problems when key pests are controlled or eliminated
114. If you have a working knowledge of the pest, you can learn about **Scouting and trapping** techniques
115. **Prevention** is an important goal
116. A very important stepin an IPM program is **Recordkeeping and evaluation** of your efforts
117. If pesticides are frequently applied during a growing season, resistance can show up **More quickly**
118. **Arthopod** serve as intermediate host and vector of pathogenic organism
119. Flies are the example of the class **Insecta**
120. The phylum is probably **Monophyletic** but with four distinct groups
121. Types of pathogens transmission are **Mechanical and biological**
122. Vector efficiency is expressed in terms of **Low high capacity**

1. What is animal science?

Refer to the total store of knowledge relative to the breeding, feeding,care and management of caring and the marketing and processing of animal and their products as gained through partial experience and research method

2. What is biosecurity?

Protection your birds from disease
Prevention or controlling disease transmission by vectors

3. What is vector?

Something that may transmit a disease
Rodents, Birds Insects, Shoes, Car tires, shared equipment, best friend, pet, food, water, dust, air etc.

Treatment of Bumbal foot

PODODERMATITIS
Injury to the lower surface of the foot and subsequent infection with*Staphylococcus bacteria*

Common causes of injury:

➢ Rough perches ➢ Splinters ➢ Wire floors ➢ Poor litter or bedding ➢ Quality

CLINICAL SIGNS

➢ Lameness ➢ Swelling of the foot pad ➢ Hard, pus-filled abscess on foot pad

TREATMENT

➢ Soak foot in warm water and Epsom salts.
➢ Disinfect with alcohol.
➢ If skin is open, drain pus from abscess.
➢ Flush abscess cavity with hydrogen peroxide (H_2O_2) to cleanout pus and debris.
➢ Pack the cavity with antibiotic ointment.
➢ Wrap the foot with gauze and elastic bandage.
➢ Repeat daily until foot heals.

PREVENTION AND CONTROL

➢ Provide good quality litter or bedding.
➢ Keep bedding clean, dry, and deep.
➢ Keep perches less than **18 inches** from the floor to prevent foot damage due to impact from jumping.
➢ Remove potential sources of injury such as sharp objects and/or surfaces.

4. Bubonic plague

✓14th Century Europe: mysterious scourge kills million.
✓Centuries later it was found that rat fleas became infected with disease-causing bacteria.
✓Fleas sought other warm-blooded host (humans) when rat numbers declined.
✓Plague is currently managed monitoring for plague and reducing the number rodent-hosts for fleas.

Natural pest control factors

➢ Wind
➢ Temperature
➢ Humidity, rain
➢ Rivers, lakes, mountains
➢ Pathogens, predators
➢ Food supply of the pest
➢ Cultural control
➢ Control by natural enemies
➢ Genetic control
➢ Natural chemical control

Natural controls are out of man's control – Mother Nature dishes these out. They are without a doubt the leading factors that manage pest populations. Natural controls include wind, temperature extremes, humidity levels, and rainfall.

A cold snap in the spring or a wet spring can largely reduce insect pest numbers for that season. Rivers, lakes, and mountains provide significant barriers to pest movement and expansion. Naturally-occurring predators and pathogens hold down pest populations. The natural food supply of the pest also affects population density.

5. Difference between monogastric and polygastric

➢ **Monogastric** (simple stomach)
o Pig, Dog, Human
➢ **Polygastric** (ruminants)
o Cow, Sheep, Goat

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6. Difference between Avain and pseudo-ruminants

➢ **Avain** (poultry) – (Gizard)

o Chicken, Turkey, Duck-

➢ **Pseudo-ruminants** (cecum)

o Horse, Rabbits, Hamster-

7. Bioaccumulation

Bioaccumulation*: when a chemical accumulates in animal fat (historical fact)*

DDT and its relative compounds were also found to be stored in fat cells of insects, fish, birds, rodents and other animals. DDT didn't break down in fat cells. This is known as bioaccumulation *than the organisms they consume*

8. Biomagnification

Bio-magnification: *when an organism accumulates residues at higher concentrations*

Some organisms accumulate chemical residues in higher concentrations than those found in their food source. This process is called **biomagnification**. For example, invertebrates with pesticides in their tissues are eaten by fish, which are then eaten by birds. The birds at the top of the food chain accumulate the highest concentration of pesticide residues. Pesticides that bioaccumulate and biomagnify have been removed from the marketplace.

9. Sources of protein 3marks.

Protein supplements:

✓Protein suppl. Added to provide the essential AAs. (Amino Acids)

✓Several protein sources used to achieve a better balance of the needed AAs.

✓Animal protein sources are more variable in their amino acids than plant protein

✓AAs req. for poultry differ from that of other animals in that glycine & serine are dietary req. & required glycine for uric acid formation

A-Plant protein:

✓Soybean meal is most commonly plant protein source & has a better balance of AAs than other plant protein (cottonseed meal, corn gluten meal, linseed meal).

✓Cottonseed meal used in grower poultry ration to replace up to 50% of the soybean meal, while linseed meal not more than 3-5% of diet.

B-Animal protein:

✓The most commonly used are fish meal, meat by-products, milk by-products, blood meal, feather meal & poultry by-product meal.

✓Fish meal have a good balance of AAs, but must not used in large amount (used at 2-5%) to avoid fishy flavor in eggs & poultry meat

10. Difference between hygiene and milk 3marks.

✓Dirty milk (containing dirt, bacteria or antibiotics) can cause serious problems when processing.

✓Dirty milk is caused by unhygienic milking machines or poor milk filters.

✓Antibiotics are found in milk as residues from treatments to cows for mastitis.

✓When milk is found to be not up to standard, it may be rejected by the creamery or bought at a lower price.

11. Milk composition 5marks

- ✓ Milk used for bottling or drinking must by law contain 3.6% fat and 8.5% SNF
- ✓ Creameries buy milk in many different ways – sometimes at flat rate per Kg.
- ✓ Sometimes, however, it may be by percentage butterfat or protein.
- ✓ Milk composition varies also amongst different breeds.
- ✓ The milk of the Jersey cow has the highest butterfat and SNF content of any cows.
- ✓ Also the composition of milk varies during milking.
- ✓ The milk at the start may only contain 1% fat, while the milk at the end ("The Stripping") may contain 10% fat

12. Common pest control factors

- ➢ Wind
- ➢ Temperature
- ➢ Humidity, rain
- ➢ Rivers, lakes, mountains
- ➢ Pathogens, predators
- ➢ Food supply of the pest
- ➢ Cultural control
- ➢ Control by natural enemies
- ➢ Genetic control
- ➢ Natural chemical control

13. 1st, 2nd, 3rd, 4th generation pesticides.

First-generation pesticides (inorganic)

– First attempt at chemical technology
– Included heavy metals such as arsenic, copper and lead.
– Toxic to humans and agricultural plants.
– Pests developed resistance

Second-generation pesticides

- ➢ Organic chemical (organochlorines).
- ➢ Used after WWII (presently in developing countries)
- ➢ Synthesis begins with petroleum ("oil")
- ➢ Mechanism of actions often unknown.
- ➢ Bioaccumulation & Biomagnification.
- ➢ Toxic to animals (humans) and agricultural plants.
- ➢ Pests developed resistance.

Third-generation Pesticides

- ➢ Organophosphates and carbamates
- ➢ Less persistent in environment (good deal)
- ➢ Acutely potent nerve toxins
- ➢ More lethal in low dose than organochlorines

Fourth-generation Pesticides

- ➢ Endocrine disruptors (hormonal chaos)
- ➢ Target a critical life cycle stage of insects.
- ➢ Not direct killers per say.
- ➢ Reduce reproduction (fertility) of population.

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14. Ecological pest management

- Control based on pest life cycle and ecology
- Control agent may be an organism or chemical
- Specific to pest and/or manipulate a part of the ecosystem
- Emphasizes protection from pest

15. Order of class insect. 5marks.

Copied from ***Google***, and have more order written then in **PPtx**. So. One can remember few.

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[Subclass] Apterygota
- Archaeognatha (Order: **Microcoryphia**)
- Three-pronged bristletails (Order: **Thysanura**)

[Subclass] Pterygota
- [Division] Exopterygota
  - Biting and Sucking lice (Order: **Phthiraptera**)
  - Booklice and Barklice (Order: **Psocoptera**)
  - Cockroaches (Order: **Blattodea**)
  - Dragonflies and Damselflies (Order: **Odonata**)
  - Earwigs (Order: **Dermaptera**)
  - Grasshoppers and Crickets (Order: **Orthoptera**)
  - Praying Mantids (Order: **Mantodea**)
  - Mayflies (Order: **Ephemeroptera**)
  - Stick insects and Leaf insects (Order: **Phasmatodea**)
  - Stoneflies (Order: **Plecoptera**)
  - Termites (Previously Order: **Isoptera** but now part of Order: Blattodea)
  - Thrips (Order: **Thysanoptera**)
  - True Bugs (Order: **Hemiptera**)
  - Web-spinners (Order: **Embioptera**)
  - Zorapterans (Order: **Zoraptera**)
- [Division] Endopterygota
  - Alderflies, Dobsonflies & Fishflies (Order: **Megaloptera**)
  - Bees, Wasps and Ants (Order: **Hymenoptera**)
  - Beetles (Order: **Coleoptera**)
  - Butterflies and Moths (Order: **Lepidoptera**)
  - Caddisflies (Order: **Trichoptera**)
  - Fleas (Order: **Siphonaptera**)
  - Flies (Order: **Diptera**)
  - Lacewings, Antlions & Mantidflies (Order: **Neuroptera**)
  - Scorpionflies (Order: **Mecoptera**)
  - Snakeflies (Order: **Raphidioptera**)
  - Strepsipterans (Order: **Strepsiptera**)

16. Mode of action of pesticides and pest resistance. (long Question)

Mode of action: how they work to control the pest **??**

– Systemic pesticides are absorbed through tissues and transported elsewhere where the pest encounters it through feeding

• Used on plants or livestock

– **Contact** pesticides must come in direct contact with the target pest

Selectivity: what range of pests they affect

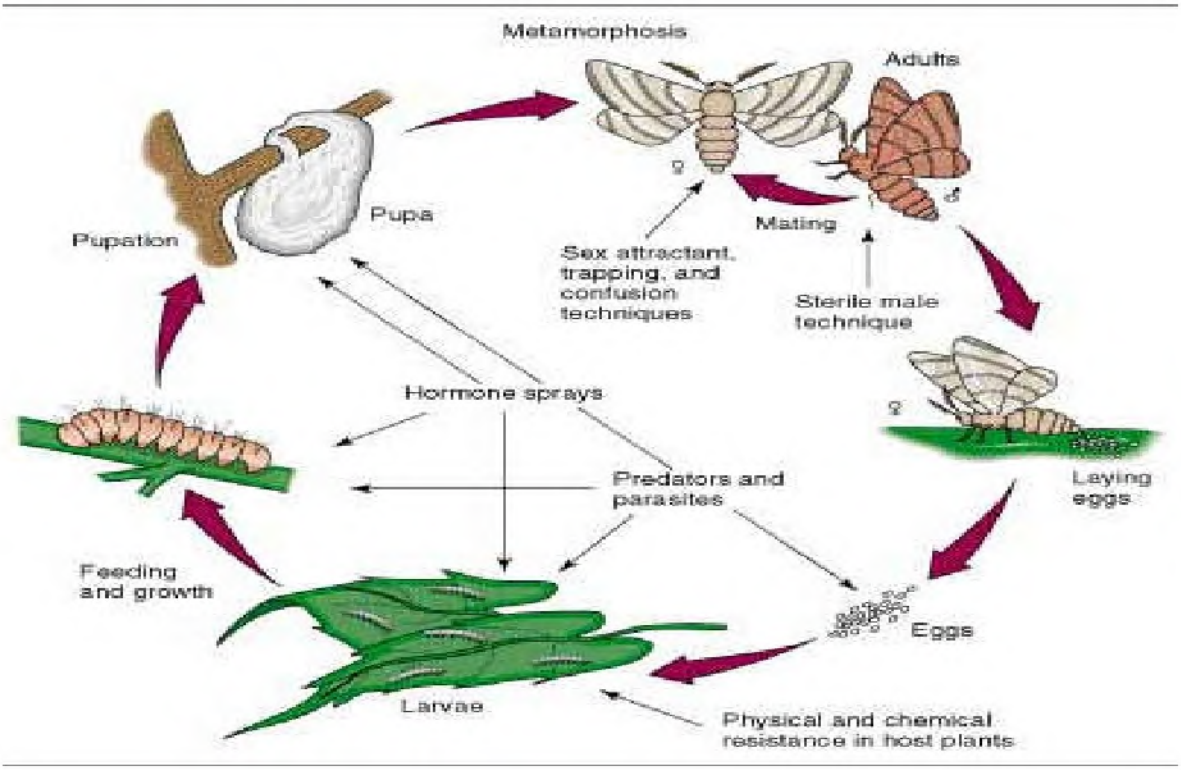
– **Non-selective –** kills all related pests – for example some herbicides kill all green plant that gets a sufficient dose

– **Selective** – kills only certain weeds, insects, plant pathogens – for example other herbicides only kill broadleaf weeds not grasses

Persistence: how long they remain active in the environment
– **Residual pesticides** – remain active for weeks, months, years – for example herbicides used around road guard rails
– **Non-residual –** inactivated immediately or within a few days – for example – some herbicides do not remain active in the soil once applied

Resistance to Pesticides
• Chemical pesticides lose effectiveness
• Resistant pest populations produce next generations
The pests had developed resistance to the chemical. Since the 1970's, pest resistance has surfaced in weed species, plant pathogens and insect pests. A prime example is the **Colorado potato beetle** which has developed resistance to every major group of insecticides, making the older chemical compounds nearly useless as pest management tools for Colorado potato beetle.

17. Insect Life cycle



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